

PROFESSIONAL SUMMARY

Medicinal chemist and computational biologist with 18 years of R&D experience spanning drug discovery, AI-driven protein design, and bioinformatics. I have a proven track record of establishing and leading an independent research group, securing competitive funding, developing higher-education curricula, and supervising 35+ HDR students and staff. Expertise in integrating machine learning and computational tools with experimental validation to accelerate the discovery and design of natural products and protein-based solutions for cancer, diabetes, neurodegeneration, and infectious diseases.

CORE COMPETENCIES

AI & Computational Biology: Machine/deep learning (Scikit-Learn, TensorFlow, PyTorch), Statistics (R/RStudio), protein structure prediction (AlphaFold and RoseTTA series) and generation (DMS, ProteinMPNN, conditioned generators), bioinformatics, NLP for biomolecules, HPC/cloud computing (AWS, Google Colab, Hugging Face).

Drug Discovery: Peptide design, medicinal chemistry, natural product chemistry, chemoinformatics, molecular docking, QSAR modelling, high-throughput screening.

Laboratory Skills: Protein expression/purification, peptide synthesis, spectroscopy (NMR, CD, UV, IR), chromatography (RP-HPLC, UPLC), mass spectrometry.

Leadership: Research strategy, grant writing (5.6M MXN secured), team supervision, curriculum development, multidisciplinary research, and cross-sector partnerships.

Languages: English (fluent, 15+ years), Spanish (fluent, 8+ years), French (native).

PROFESSIONAL EXPERIENCE

Founder & Lead Consultant, Ingenie Bio – Sydney, Australia | Oct 2024-present

- Delivering AI/ML advisory services across sectors to accelerate molecular discovery and optimisation.
- Implementing advanced models for property/developability prediction and candidate prioritisation.
- Designing computational pipelines to support rational protein engineering and reducing failure rates.

Group Leader (Senior Lecturer), CINVESTAV-IPN – Irapuato, Mexico | Nov 2017-Jun 2024

- Led a computational biology research laboratory advancing AI-driven peptide therapeutics, establishing integrated in silico pipelines for bioactive candidate discovery and optimisation.
- Supervised 17 students (5 BSc, 8 MSc, 4 PhD) and mentored 18 additional HDR researchers, contributing to regional talent development in molecular sciences.
- Secured 5.6 million MXN in competitive research funding despite systemic R&D underinvestment, sustaining multi-year research programs and laboratory expansion.
- Co-invented predictive and generative AI models leveraging sequence, structure, and evolutionary data to enable data-driven peptide engineering.
- Implemented structural bioinformatics and protein design workflows using AlphaFold, RoseTTA or ProteinMPNN for structure-guided engineering initiatives.
- Conducted high-throughput venom transcriptome analysis *via* HPC, machine learning, and phylogenetics to expand bioactive peptide discovery space.
- Designed and optimised protein expression and purification protocols for small proteins to support downstream functional and structural validation.
- Partnered with experimental teams to validate bioactive peptides targeting membranes and receptors, syncing computational predictions and bioactivity assays.
- Delivered courses in data science, chemical biology, chemoinformatics, and structural bioinformatics, embedding computational methods in postgraduate curricula.
- Initiated industry collaborations with *Mexican citrus growers* and *Nature Sweet*, applying peptide discovery platforms to agricultural pathogen pests.

Health Data Scientist Fellow, Insight Data – Boston, USA | Sept 2016-Feb 2017

- Developed ML models predicting adverse immunogenic reactions using >450,000 vaccine reports.
- Led full-stack software development to deploy [Reac2Vac.online](#) for public access.
- Architected and operated a cloud-based predictive analytics platform on Amazon Web Services.

Research Officer, The University of Queensland – Brisbane, Australia | Sept 2012-Sept 2015

- Co-developed novel peptide therapeutics for type 2 diabetes in collaboration with **Pfizer**, advancing GLP-1 receptor targeting strategies.
- Engineered and experimentally validated GLP-1 receptor agonists demonstrating potency comparable to Exenatide (Byetta™).
- Developed protein-protein interaction (PPI) inhibitors targeting myeloid leukemia in partnership with **Protagonist Therapeutics**.
- Applied Schrödinger Maestro to conduct conformational exploration and design for structure-informed optimization of therapeutic macrocycles and peptides.

PhD Researcher, UQ Institute for Molecular Bioscience – Brisbane, Australia | Aug 2008-Aug 2012

- Discovered marine-derived kinase inhibitors in collaboration with Spanish biotech **Noscira**, contributing to novel molecular entities in oncology and neurodegeneration.
- Led structure-based drug discovery, generating 5 highly potent pM kinase inhibitors (**EP2662081A**) with target selectivity.
- Optimised purification and structural characterisation studies of scarce marine natural products (0.1-5mg), leading to the identification of 30+ accurate NMR and MS analyses.
- Developed a predictive ML model for blood-brain permeability, accelerating candidate prioritization in CNS drug discovery.
- Contributed to high-throughput screening (HTS) campaigns for antimalarial discovery in collaboration with the **QIMR Berghofer MRI**, identifying promising lead compounds.

Research Engineer, PIERRE FABRE and CNRS – Toulouse, France | Mar 2007-May 2008

- Synthesised 100+ analogues from limited natural products (5-10mg scale) under sensitive conditions, expanding chemical diversity for bioactivity testing.
- Co-invented a patented synthetic process (**FR2941697A1**), streamlining multi-step chemical synthesis and improving overall yield and efficiency.

EDUCATION & TRAINING

2012 **Ph.D. Organic Chemistry**, The University of Queensland (UQ) – Australia

2007 **MSc Chemical Engineering**, ENSCM, Grande École – Montpellier, France

2007 **MSc Biomolecular Engineering**, ENSCM, Grande École – Montpellier, France

SELECTED PUBLICATIONS & PATENTS

26 peer-reviewed publications in *Springer Nature Scientific Reports*, *Chemical Science*, *Angewandte Chemie*, *ChemBioChem*, *Digital Discovery*, *Microbial Genomics*, and others.

- Aguilera-Puga, M.d.C. & Plisson, F.* Structure-aware machine learning strategies for antimicrobial peptide discovery. *Scientific Reports* 2024, 14, 11995. [\[DOI\]](#) [\[Research Square\]](#) [\[Github\]](#)
- Saldívar-González, F. I.; Aldas-Bulos, V. D.; Medina-Franco, J. L. & Plisson, F.* Natural Product Drug Discovery in the Artificial Intelligence Era. *Chemical Science* 2022, 13, 1526-1546. [\[DOI\]](#)
- Plisson, F.*; Ramírez-Sánchez, O. & Martínez-Hernández, C. Machine learning-guided discovery and design of non-hemolytic peptides. *Scientific Reports* 2020 10, 16581. [\[DOI\]](#) [\[Github\]](#)
- Plisson, F.; Hill, T. A.; Mitchell, J. M.; et al. Helix-constraints and amino acid substitution in GLP-1 increase cAMP and insulin secretion but not β -arrestin 2 signalling. *European Journal of Medicinal Chemistry* 2017, 127, 703-71. [\[DOI\]](#)

1 patent: Derguini, F.; Plisson, F.; Massiot, G. Preparation of tagitinin C and F derivatives as anti-cancer agents. Patent application FR 2941697 A1 20100806 (2010) [\[Europe PMC\]](#) [\[Google Patents\]](#).

GRANTS & AWARDS

Lead CI/PI on 5.6 million MXN in grants from CONAHCYT, FOSEC SEP, and CGSTIC (2017-2024).

Roche & Mexican Health Foundation - Rosenkranz Medical Research Prize in Biotechnology (2021).

Forum for Health Research from Guanajuato State Award (2018).

Mexican Research Council Early-Career Research Fellowship (2017-2022).

IMB Research PhD Scholarship & UQ International Tuition Research Award (2008-2012).

SELECTED PRESENTATIONS & SERVICES

- ▷ *Scientific outreach*: 65+ seminars at universities and conferences across the Americas, Europe, Asia, and Australia (2017-2025), 15 conference proceedings and two podcasts.
- ▷ *Panel discussions*: Symposium on Bioinformatics and Engineered Intelligence (Adelaide, 2025), Hit ID Symposium (Melbourne, 2025), Health Data Science (online, 2021), Allbiotech Summit (Mexico, 2018)
- ▷ *Workshop organiser/speaker*: BIRS-CMO Computational Biology (Oaxaca, 2023), RIIAA (Mexico City, 2021/2022), CABANA Chemoinformatics (Mexico, 2019) & Bioinformatics (Colombia, 2019)
- ▷ *Peer-reviewer*: 60+ manuscripts for Nature, RSC, ACS, Oxford, BMC journals.
- ▷ *Grant reviewer*: Mexican Research Council, Canada NFRF, UK Rosetree Trust, European Commission.
- ▷ *External PhD examiner*: University of Queensland (2021), Tecnológico de Monterrey (2021), University of Guanajuato (2018).

CONTINUOUS LEARNING

- 2024 mRNAs as Medicines (Moderna)
- 2024 Machine learning in production - MLOps (deeplearning.ai)
- 2024 Open Source Models with Hugging Face (deeplearning.ai)
- 2023 How Diffusion Models Work (OpenAI & deeplearning.ai)
- 2022 Deep Learning for Molecules and Materials (A.D. White, dmol.pub)
- 2022 STAT 453: Introduction to Deep Learning (S. Raschka, sebastianraschka.com)
- 2021 MIT 6.S191: Introduction to Deep Learning (introtodeeplearning.com)
- 2020 Sequence models (Deep Learning Specialisation) (A. Ng, deeplearning.ai)
- 2019 Neural Networks and Deep Learning (A. Ng, deeplearning.ai)
- 2013 Data Analysis, Johns Hopkins Bloomberg School of Public Health (J. Leek)